

Literature status: finite-dimensional logmodular algebras

Run `fa_banach_001` — lane 12

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Status. *Literature already answered.* This is an exact later theorem, not a new run proof.

1 Source question

Paulsen and Raghupathi, *Representations of Logmodular Algebras* (arXiv:0806.0284), prove in Theorem 2.2 that a logmodular subalgebra $\mathcal{A} \subseteq M_n$ containing the diagonal is, after permuting the basis, a block upper-triangular algebra. Immediately after Corollary 2.3 on PDF page 4 they state that they do not know whether every logmodular subalgebra of M_n has this form, but believe that it does.

Equivalently, the question is whether the diagonal-containment hypothesis can be removed, up to unitary equivalence.

2 Exact later theorem

Juschenko’s *A description of the logmodular subalgebras in the finite dimensional C^* -algebras* (arXiv:1003.3123) explicitly says in its abstract and introduction that it proves the conjecture from Paulsen–Raghupathi. Its main result is:

Theorem 2.6 (Juschenko, PDF page 5). If $\mathcal{A} \subseteq M_n(\mathbb{C})$ is logmodular, then $\mathcal{A}$ is unitarily equivalent to an algebra of block upper-triangular matrices.

This is precisely the missing classification. The supporting paper then deduces in Corollary 2.7 that every contractive unital homomorphism from such an algebra into $B(H)$ is completely contractive.

The supporting author knew she was resolving the source conjecture: the abstract says the classification “was conjectured” in the cited Paulsen–Raghupathi paper, and the introduction repeats that attribution before announcing Theorem 2.6.

3 Scope and search record

The four run indexes and the parsed corpus contained no earlier packet for arXiv:0806.0284. Exact arXiv searches for logmodular subalgebras of M_n and block upper-triangular classification returned arXiv:1003.3123 as a direct answer. Both source PDFs were inspected locally.

This result is finite-dimensional. It does not settle the source’s separate questions about $H^\infty(\mathbb{D})$, infinite upper-triangular operator algebras, completely distributive CSL algebras, or the final abstract operator algebra $\mathcal{C}$.

References

1. V. I. Paulsen and M. Raghupathi, *Representations of Logmodular Algebras*, Trans. Amer. Math. Soc. 363 (2011), 2627–2640; arXiv:0806.0284.
2. K. Juschenko, *A description of the logmodular subalgebras in the finite dimensional C^* -algebras*, arXiv:1003.3123, Theorem 2.6.